

One Prediction, Three Voices
Designing AI That Explains Itself Differently to Different People
Maitreya Sapariya
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Section 1: Problem Statement

An AI system that predicts preeclampsia risk from routine antenatal data faces a problem that has nothing to do with the accuracy of its predictions: a single explanation, delivered the same way to everyone, works for whichever user it was designed around and quietly fails everyone else. In rural India, three people typically stand between a preeclampsia-risk model and a clinical decision: the Accredited Social Health Activist (ASHA) who visits the household, the pregnant woman whose risk is being assessed, and the clinician who acts on a referral. Roughly 1.05 million ASHAs work across India (Ministry of Health and Family Welfare, 2025), most with a formal-education floor of Class 8 or 10. The women they serve face a documented digital gap: rural computer literacy stood at 18.1% against 39.6% urban (National Statistical Office, 2021), and only 33.9% of rural women had used the internet, against 55.6% of rural men (National Family Health Survey-5, 2019–21). A visual dashboard, or a SHAP chart plotting feature contributions, assumes a level of digital and chart literacy that a large share of this population does not have.

The barrier is not one problem but two that get conflated. Digital literacy and sensory ability are separate axes: a person can read fluently and still be excluded by an interface built around comfortable screen use, just as a fully digital-literate person can be excluded by a cluttered chart if their vision is impaired. Layered onto this is a cognitive load problem independent of literacy: interpreting a probabilistic risk score, understanding that a 68% estimate is neither a diagnosis nor a guarantee, is demanding even for a literate, numerate reader. A visual chart adds a second translation step for anyone not already comfortable reading charts.

India's Rights of Persons with Disabilities Act (2016), through Sections 40, 42, and 46, mandates ICT accessibility and points to WCAG 2.1 compliance via the national standard IS 17802. That framework is built around twenty-one recognized disability categories. It was not written to address a gap in digital or health literacy that has nothing to do with disability status. A mother-to-be who is fully sighted and hearing but has never used a smartphone falls

outside what current accessibility law is designed to catch, though she is just as excluded as a user the law does recognize. That gap, between who accessibility law protects and who a real deployment actually excludes, is the specific problem this project addresses.

Section 2: Proposed Solution

The proposed system is a single prediction model with three explanation layers, not three separate systems. The model is planned as a comparison across Logistic Regression, Random Forest, and XGBoost, trained on features an ASHA can collect during a routine home visit without lab equipment: blood pressure, proteinuria, edema, and obstetric history. SHAP value attribution identifies which features drove a given prediction. Training uses stratified k-fold cross-validation, with SMOTE applied only within training folds to correct class imbalance. The primary target is binary, preeclampsia versus normal; a three-class severity distinction is secondary and exploratory. No results exist yet.

For the clinician, the output is the full SHAP dashboard: every feature, its contribution, and its direction, presented the way a trained reader expects a technical decision-support tool to look. No simplification, because none is needed.

For the ASHA worker, the instinct runs the other way: narrower, not more detailed. The output surfaces the two or three factors that mattered most, in plain language, plus one recommended next step, such as a referral within a given timeframe. Scott et al. (2022) found frontline health workers juggle overlapping paper and digital records with little incentive to keep them accurate, since oversight rewards completeness over accuracy. Adding a fourth register to log would repeat that mistake; the goal here is a lighter cognitive load, distinct from simply shortening the clinician's dashboard.

For the mother-to-be, the explanation is spoken, in Hindi, not written. She hears what the risk means and what to do next without needing to be a confident reader first. This is not a workaround for a disability; it follows from Gaver's (1991) argument that an affordance can be carried across different perceptual channels, the same information a chart shows visually, a voice channel can carry aurally, as long as the substitution matches how she actually receives information.

Designing for the mother-to-be's tier first, rather than the clinician's, was the harder constraint, and solving it is what forces the tiered architecture to exist at all: starting from the tier with the least room for error extends more easily upward to the ASHA and clinician tiers than the reverse.

Across all three tiers, the model's role stays the same: it flags risk and surfaces the factors behind that flag. It does not issue a diagnosis or make the referral decision; that stays with the ASHA or the clinician. The system increases what each user can act on, not the judgment they already exercise. The ASHA tier's shift from more detail to less came directly from conversations with a clinical collaborator during this project's design phase, grounded in her feedback rather than assumed in advance, and the same collaborator feeds the evaluation plan: clinician feedback plus a heuristic walkthrough for the ASHA and mother-to-be tiers, checking the design against real users before it is trusted.

Section 3: Evidence Base

The design in Section 2 draws on two bodies of evidence that rarely cite each other: inclusive design theory, and the actual track record of voice-based health messaging in India.

Three frameworks anchor the tiered approach. The Web Content Accessibility Guidelines organize accessibility around four principles: information must be perceivable, operable, understandable, and robust across different ways of receiving it. Universal Design (Mace et al., 1997) argues for one system serving a wide range of users through flexible interaction, rather than a separate system per group, the same logic behind treating this as one model with three explanation layers instead of three products. Ability-Based Design (Wobbrock et al., 2011) reframes the question further: instead of what a user lacks, it asks what they can already do, and designs around that. The ASHA and mother-to-be tiers, applied here, are not reduced versions of the clinician's tier; they are built around what each user is already fluent in, not what they are missing.

Gaver's (1991) sensory-substitution argument, already the basis for the voice tier in Section 2, has a limit worth stating plainly. The WHO's World Report on Vision (2019) estimates at least 2.2 billion people live with some vision impairment, and the World Report on Hearing (2021) puts hearing loss above 1.5 billion, with over 430 million requiring rehabilitation. A voice-first design does not reach a mother-to-be who also has significant hearing loss. India's National Blindness and Visual Impairment Survey (Vashist et al., 2022) found illiteracy strongly associated with blindness (adjusted OR 5.6), meaning the two barriers treated as separate axes in Section 1 often compound in the same population. Voice solves the literacy problem. It does not solve every accessibility problem, and this report does not claim it does.

The clinical evidence for voice-based maternal messaging in India is genuinely mixed, worth stating honestly rather than citing only the encouraging half. Kilkari, India's largest

maternal IVR program, was evaluated in a randomized controlled trial (LeFevre et al., 2022) that found no significant effect on its primary outcome, exclusive breastfeeding, alongside modest gains in immunization and contraceptive uptake. mMitra (Murthy et al., 2020) showed real gains in delivery-related practices but not in knowledge, and its primary anemia outcome could not be assessed since hemoglobin data was missing from most records. A restless-bandit trial by ARMMAN and Google AI (Mate et al., 2022) reported a 32% reduction in engagement drops that did not replicate later. None of this argues against voice as a channel, which has a strong foundation in low-literacy contexts (Patel et al., 2010). It argues for building the evaluation plan in Section 2 around what voice actually changes for this tool, rather than assuming the channel guarantees the outcome.

Section 4: Implementation Considerations and Limitations

Four limitations sit between this design and anything closer to real deployment.

The first is infrastructure, not explanation design. ANMOL, the existing ASHA-facing data-entry app tied to the RCH Portal, already supports offline entry with delayed sync rather than requiring a constant connection. A model requiring continuous network access would be unusable in the same field conditions the ASHA tier is meant to lighten, so any implementation must match that existing offline-first pattern.

The second is adoption risk, separate from whether the tool works technically. Scott et al. (2022) found that oversight of ASHA-maintained records already rewards completeness over accuracy, a worker is checked on whether a field is filled in, not whether the number in it is right. A tool that adds a step without removing an existing one risks becoming exactly what it was designed to avoid.

The third is cost, which this report has not fully worked out. Maintaining a Hindi voice layer, and extending it to other Indian languages if this moved beyond one state, requires ongoing translation and audio production, not a one-time cost. Model hosting and the clinician dashboard add a smaller, more predictable expense. A full cost estimate remains an open question here, an honest gap rather than a hidden one.

The fourth is the edge case named in Section 3: a voice-first design does not reach a mother-to-be who also has significant hearing loss, and this proposal does not currently have an answer for her.

Bringing this closer to real use needs more than an engineering team: the ASHA workers and mothers-to-be the tiers are built for, already named in the evaluation plan in Section 2; a clinical collaborator involved in this project's design; and an ethics review body,

since no version of this tool should be tested against real pregnant women's data without that clearance first.

Section 5: References

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